Asim Mahat

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Technical Skills

Languages: Python, Go, SQL, C/C++, JavaScript, HTML/CSS, Bash Scripting
Frameworks: TensorFlow, PyTorch, Scikit-learn, OpenCV, React, Node.js, Django, FastAPI, LangChain, LangGraph
Developer Tools: Git, Docker, Airflow, AWS, Jenkins, Kafka, Kubernetes, Azure, Jira, Pinecone, Qdrant, FAISS
Libraries: Pandas, Numpy, Matplotlib
Certification: Machine Learning, Deep Learning

Experience

Graduate Teaching/Research Assistant — *Wichita, KS, USA*
Wichita State University — *Aug 2024 – Dec 2025*

- Conducting research on NextG wireless communication, designing ML-based models that improved network coverage prediction accuracy by 15% and enhanced resource allocation efficiency by 20%, validated across 1,000+ simulated network scenarios
- Using HPC clusters to accelerate large-scale model training and deployment, supporting 50+ experiments weekly and cutting model evaluation cycles from days to hours
- Designing and implementing LangGraph-driven AI Agent ecosystems with FastAPI, transitioning from a monolithic architecture to scalable microservices. Achieved a 50% reduction in latency and a 30% increase in system throughput for workflows used by 2,500+ active users
- Mentored 50+ students in Operating Systems and directed 20+ senior design teams, ensuring 100% project completion rate

Software Engineer — *Kathmandu, Nepal*
Cotiviti — *Feb 2023 – Aug 2024*

- Led research initiatives to develop machine learning models aimed at detecting anomalies and fraudulent claims within US healthcare datasets, enhancing detection accuracy by 30%
- Maintained and optimized REST APIs using Django Rest Framework, managing 15+ database models and 30+ serializers/views to handle 50K+ requests/day with sub-second response times
- Collaborated with cross-functional teams, including data scientists and healthcare professionals, to design and implement AI solutions that comply with **HIPAA** regulations
- Implemented Docker to containerize applications, ensuring consistent and reproducible environments
- Optimized SQL stored procedures, views, and DB objects, reducing average query response time by 25% across core analytics pipelines

Software Engineer — *Lalitpur, Nepal*
Code Himalaya Pvt Ltd — *June 2022 – Dec 2022*

- Developed and deployed RESTful APIs using Django Rest Framework and Docker, improving backend scalability and deployment speed by 40%
- Contributed to the eKYC module of a fintech application, automating user verification and reducing onboarding time by 60%, enhancing customer experience
- Diagnosed and resolved production issues in staging and live environments using Sentry, decreasing system errors by 95% and boosting user satisfaction scores by 30%
- Optimized legacy database queries, reducing query execution time by 70% and increasing overall application performance by 45%

Software Engineer Intern — *Lalitpur, Nepal*
LIS Nepal Pvt Ltd — *Jan 2022 – Apr 2022*

- Developed Python scripts for ETL/ELT workflows, automating data transformation and ensuring end-to-end data integrity
- Built and optimized Azure Data Factory pipelines to load data into Snowflake, reducing data load time by 80%
- Monitored data pipelines in Azure and resolved SQL-based batch processing errors
- Supported delivery of a large-scale data warehouse for retail clients, contributing to data modeling and pipeline setup in Snowflake

Education

Wichita State University — *Wichita, KS, USA*
Masters of Science in Computer Science — *Aug 2024 – May 2026*

Kathmandu University — *Kavre, Nepal*
Bachelors of Science in Computer Science — *Sept 2017 – May 2022*

Publications & Contributions

- Paper titled "NextG Base Station Placement and Failure Mitigation" accepted for **IEEE ICNC** (International Conference on Computing, Networking and Communications) (**Acceptance Rate: 27%**) 2026, happening at Maui, Hawaii
- Poster: "LSTM-Based Stock Price Forecasting with PSO and GA Optimization" – Presented at Kansas Data Science Consortium (KDSC), 2025.
- Open Source Contributor at safe-mcp

Projects

- Zero Shot Traffic Signal Control - Engineered an adaptive traffic signal control agent using Llama 3 (via Ollama) for zero-shot reasoning, reducing average intersection wait times by 35% in simulation through real-time decision-making without task-specific training
- Stock Market Prediction - Achieved 92% directional accuracy predicting stock trends using LSTM. Work selected for poster presentation at KDSC 2025; full research paper currently in preparation